

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Mathematics for Physics - I
Credits: 3

L-T-P: 3-0-0

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Curvilinear coordinates, Cartesian, Spherical Polar, Cylindrical, Parabolic coordinates, Tensors, Contravariant and Covariant vectors and tensors, Mixed tensors, Addition, Multiplication and Contraction, Invariants, Differentiation of tensors. (10)
Complex analysis: Cauchy-Riemann conditions; Cauchy's theorem; Taylor series and Laurent series; zeros and isolated singularities of analytic functions; Calculus of residues; Multivalued functions and Riemann surfaces; analytic continuation. (14)
Function spaces, space of continuous functions; metric properties of the space of continuous functions; expansions in orthogonal functions; Hilbert space; Weierstrass theorem; orthogonal polynomials-Generalised Rodriguez formula; differential equations satisfied by classical polynomials; (10)
Ordinary differential equations, second order differential equations; the Sturm-Liouville problem; The Hypergeometric and related functions. The Confluent Hypergeometric and the related functions. (8)

Reference Books:

1. Mathematics for Physicists: P. Denner and A. Krzywicki
2. Methods of Mathematical Physics: R. Courant and D. Hilbert

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Curvilinear coordinates, Cartesian, Spherical Polar, Cylindrical, Parabolic coordinates, Tensors, Contravariant and Covariant vectors and tensors, Mixed tensors, Addition, Multiplication and Contraction, Invariants, Differentiation of tensors.	10
2.	Complex analysis: Cauchy-Riemann conditions; Cauchy's theorem; Taylor series and Laurent series; zeros and isolated singularities of analytic functions; Calculus of residues; Multivalued functions and Riemann surfaces; analytic continuation.	14
3.	Function spaces, space of continuous functions;	10

	metric properties of the space of continuous functions; expansions in orthogonal functions;. Hilbert space; Weierstrass theorem; orthogonal polynomials-Generalised Rodriguez formula; differential equations satisfied by classical polynomials	
4.	Ordinary differential equations, second order differential equations; the Sturm-Liouville problem; The Hypergeometric and related functions. The Confluent Hypergeometric and the related functions	8
Total number of hours		42